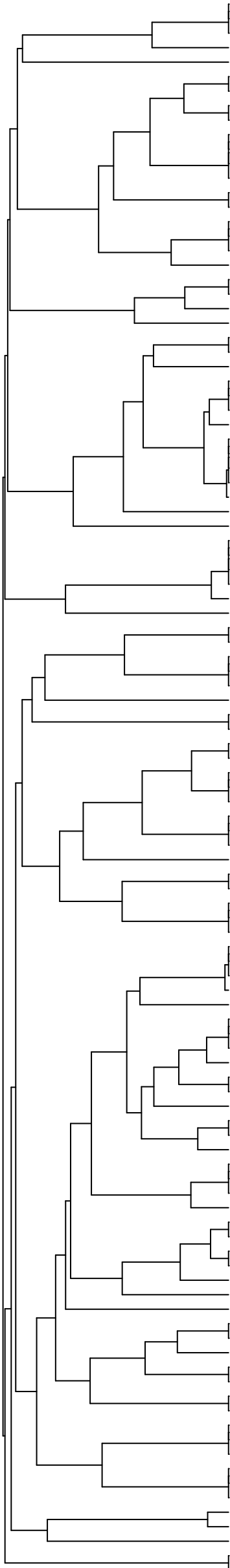


9/86 protein localization to endoplasmic reticulum
36/560 protein localization to organelle
24/312 establishment of protein localization to organelle
9/98 protein targeting to membrane
21/249 protein targeting
8/63 cotranslational protein targeting to membrane
48/791 peptide transport
14/164 establishment of protein localization to membrane
28/386 localization within membrane
2/57 protein localization to mitochondrion
42/423 amide biosynthetic process
9/107 translational initiation
24/310 peptide biosynthetic process
9/89 cytoplasmic translation
30/394 peptide metabolic process
54/615 amide metabolic process
15/179 mRNA catabolic process
7/80 nuclear-transcribed mRNA catabolic process, nonsense-mediated decay
56/762 macromolecule catabolic process
50/707 RNA processing
0/14 spliceosomal tri-snRNP complex assembly
16/217 protein-RNA complex organization
43/544 mRNA metabolic process
3/26 termination of RNA polymerase II transcription
33/389 ribonucleoprotein complex biogenesis
1/17 ribosomal small subunit assembly
23/244 ribosome biogenesis
4/49 ribosome assembly
15/195 rRNA metabolic process
3/14 folic acid metabolic process
33/481 protein catabolic process
19/259 proteasomal protein catabolic process
6/75 ERAD pathway
1/11 endoplasmic reticulum mannose trimming
28/400 proteolysis involved in protein catabolic process
20/216 response to endoplasmic reticulum stress
3/21 protein exit from endoplasmic reticulum
13/79 endoplasmic reticulum unfolded protein response
18/112 response to unfolded protein
20/133 response to topologically incorrect protein
9/43 IRE1-mediated unfolded protein response
16/160 protein folding
4/17 obsolete protein mannosylation
2/26 positive regulation of DNA-templated transcription initiation
3/15 cellular response to exogenous dsRNA
5/66 roof of mouth development
35/654 regulation of anatomical structure morphogenesis
2/24 face morphogenesis
2/34 head morphogenesis
2/35 face development
4/48 cardiac ventricle morphogenesis
7/103 cardiac chamber development
5/74 cardiac ventricle development
24/415 heart development
4/51 artery development
18/293 vasculature development
31/569 circulatory system development
360 muscle organ morphogenesis
16/242 muscle organ development
26/431 muscle structure development
13/220 muscle tissue development
0/14 neural plate pattern specification
22/370 sensory organ morphogenesis
7/158 mesoderm development
3/90 mesoderm morphogenesis
4/119 formation of primary germ layer
0/11 imaginal disc lineage restriction
4/76 imaginal disc pattern formation
14/396 cell fate commitment
48/729 tube morphogenesis
2/81 leg disc development
49/673 regulation of locomotion
4/59 morphogenesis of an epithelial sheet
52/795 cell morphogenesis
15/285 neuron projection guidance
29/432 axon development
14/259 regulation of Wnt signaling pathway
2/29 regulation of epidermal cell differentiation
4/70 actin filament-based movement
25/369 actin filament-based process
3/28 neuromuscular process controlling balance
35/641 mitotic cell cycle
1/10 mitotic nuclear membrane disassembly
1/13 spindle midzone assembly
40/757 cytoskeleton organization
28/519 protein phosphorylation
21/183 regulation of small GTPase mediated signal transduction
27/240 regulation of GTPase
21/11 regulation of vascular permeability
22/242 circulatory system process
24/338 regulation of system process
2/55 regulation of heart rate
9/151 regulation of blood circulation
10/137 muscle system process
12/291 regulation of trans-synaptic signaling
2/35 regulation of synaptic transmission, cholinergic
3/80 regulation of amine transport
33/625 cell-cell signaling
27/392 regulation of hormone levels
6/128 response to ammonium ion
12/206 reproductive behavior
1/35 short-term memory
5/163 memory
8/161 learning
15/318 cognition
6/116 associative learning
7/87 olfactory behavior
10/133 chemosensory behavior
12/265 locomotory behavior
27/473 regulation of anatomical structure size
12/148 regulation of tube size
4/29 intracellular monoatomic anion homeostasis
7/83 response to fatty acid
25/282 response to acid chemical
20/265 response to alcohol
2/35 behavioral response to ethanol
3/34 negative regulation of G protein-coupled receptor signaling pathway
5/79 regulation of G protein-coupled receptor signaling pathway
2/23 adaptation of signaling pathway
24/305 G protein-coupled receptor signaling pathway
0/17 negative regulation of DNA-templated transcription, elongation
1/15 response to anoxia
4/29 intracellular transport involved in cilium assembly
6/90 oxidative phosphorylation
22/902 generation of precursor metabolites and energy
3/26 mitochondrial electron transport, NADH to ubiquinone
14/137 cellular respiration
15/197 energy derivation by oxidation of organic compounds
4/54 mitochondrial ATP synthesis coupled electron transport
9/125 electron transport chain
0/14 energy coupled proton transport, down electrochemical gradient
17/192 nucleoside monophosphate metabolic process
14/177 nucleoside triphosphate metabolic process
7/35 NADH dehydrogenase complex assembly
21/335 mitochondrion organization
8/68 mitochondrial respiratory chain complex assembly
2/15 cAMP metabolic process
30/343 ribose phosphate metabolic process
3/70 proton transmembrane transport

p < 0.001
p < 0.01
p < 0.05

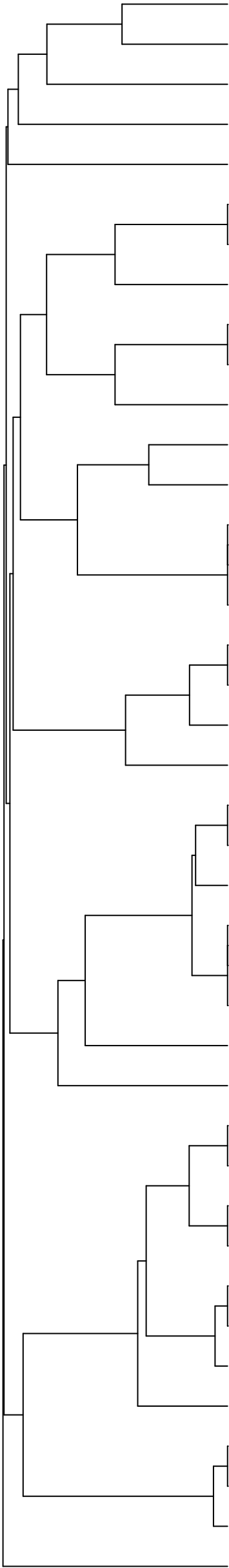
SUMMER – Biological Process (BP)



- 1/11 oligosaccharyltransferase complex
- 70/767 obsolete endoplasmic reticulum part
- 12/95 endoplasmic reticulum protein-containing complex
- 3/12 endoplasmic reticulum chaperone complex
- 3/22 COP1-coated vesicle
- 17/172 obsolete cytosolic part
- 13/92 cytosolic ribosome
- 5/33 cytosolic small ribosomal subunit
- 5/54 small ribosomal subunit
- 9/86 large ribosomal subunit
- 18/159 ribosome
- 8/54 cytosolic large ribosomal subunit
- 50/675 ribonucleoprotein complex
- 5/23 obsolete polysomal ribosome
- 6/60 obsolete polysome
- 5/24 rough endoplasmic reticulum membrane
- 9/73 rough endoplasmic reticulum
- 4/14 cytoplasmic side of rough endoplasmic reticulum membrane
- 5/24 cytoplasmic side of endoplasmic reticulum membrane
- 5/82 Sm-like protein family complex
- 1/21 U5 snRNP
- 2/28 spliceosomal tri-snRNP complex
- 1/13 SMN-Sm protein complex
- 49/682 obsolete mitochondrial part
- 29/342 mitochondrial matrix
- 11/78 oxidoreductase complex
- 0/11 proton-transporting ATP synthase complex
- 23/321 organelle inner membrane
- 5/78 inner mitochondrial membrane protein complex
- 9/175 mitochondrial protein-containing complex
- 6/59 respiratory chain complex
- 32/463 mitochondrial envelope
- 4/26 NADH dehydrogenase complex
- 50/746 organelle envelope
- 10/141 obsolete mitochondrial membrane part
- 1/18 cytochrome complex
- 2/29 proton-transporting two-sector ATPase complex
- 1/21 proteasome regulatory particle
- 4/51 proteasome complex
- 0/11 proteasome regulatory particle, base subcomplex
- 5/82 peptidase complex
- 1/10 cytosolic proteasome complex
- 7/134 ficolin-1-rich granule
- 12/156 contractile muscle fiber
- 0/10 myofilament
- 14/212 actin cytoskeleton
- 2/13 myosin II complex
- 5/29 myosin complex
- 3/11 female germline ring canal
- 4/56 cleavage furrow
- 5/80 cell division site
- 1/14 mitotic spindle midzone
- 2/41 spindle midzone
- 5/55 spindle microtubule
- 13/235 spindle
- 57/761 microtubule cytoskeleton
- 2/19 microtubule end
- 13/138 microtubule
- 28/356 supramolecular fiber
- 2/45 kinesin complex
- 8/226 chromosomal region
- 0/16 chromocenter
- 8/128 condensed chromosome
- 2/38 condensed chromosome, centromeric region
- 1/12 outer kinetochore
- 8/297 obsolete axon part
- 19/482 axon
- 5/100 terminal bouton
- 6/148 neuron projection terminus
- 0/46 main axon
- 26/577 synapse
- 4/69 presynaptic membrane
- 11/147 synaptic membrane
- 16/310 presynapse
- 13/265 postsynapse
- 7/106 postsynaptic membrane
- 4/42 neuronal cell body membrane
- 7/131 neuron spine
- 23/441 dendritic tree
- 1/56 dendritic shaft
- 7/96 leading edge membrane
- 12/239 cell leading edge
- 3/45 neuron projection membrane
- 14/189 cell projection membrane
- 5/70 plasma membrane signaling receptor complex
- 10/133 receptor complex
- 54/630 obsolete intrinsic component of plasma membrane
- 0/24 neurotransmitter receptor complex
- 10/215 plasma membrane protein complex
- 3/110 monoatomic ion channel complex
- 34/325 cell surface
- 40/582 plasma membrane region
- 15/163 basolateral plasma membrane
- 4/65 basal part of cell
- 18/317 apical part of cell
- 10/205 apical plasma membrane
- 14/130 cluster of actin-based cell projections
- 13/97 brush border
- 4/72 apical junction complex
- 4/34 bicellular tight junction
- 4/50 tight junction
- 7/148 adherens junction
- 14/290 anchoring junction
- 1/16 muscle tendon junction
- 4/36 ciliary tip
- 3/22 intraciliary transport particle
- 3/19 dendrite terminus
- 3/62 protein kinase complex
- 10/213 transferase complex, transferring phosphorus-containing groups

p < 0.001
p < 0.01
p < 0.05

SUMMER – Cellular Component (CC)



4/32 tumor necrosis factor receptor superfamily binding

22/260 ubiquitin-like protein ligase binding

3/66 ubiquitin-like protein binding

4/79 unfolded protein binding

26/502 DNA-binding transcription factor

21/314 structural molecule

12/124 structural constituent of ribosome

2/28 structural constituent of muscle

1/46 translation initiation factor

52/679 RNA binding

3/43 rRNA binding

1/20 plus-end-directed microtubule motor

9/126 microtubule binding

28/468 cytoskeletal protein binding

8/160 actin binding

4/74 actin filament binding

29/317 nucleoside-triphosphatase regulator

19/162 guanyl-nucleotide exchange factor

29/375 GTPase binding

9/122 GTPase activator

1/70 ligand-gated monoatomic cation channel

6/163 monoatomic cation channel

8/102 ligand-gated monoatomic ion channel

3/74 voltage-gated channel

17/235 channel

11/152 gated channel

3/49 neurotransmitter receptor

5/70 channel regulator

11/78 active monoatomic ion transmembrane transporter

12/125 primary active transmembrane transporter

5/38 oxidoreduction-driven active transmembrane transporter

8/83 proton transmembrane transporter

6/79 oxidoreductase, acting on NAD(P)H

4/27 NADH dehydrogenase

4/34 oxidoreductase, acting on NAD(P)H, quinone or similar compound as acceptor

9/89 electron transfer

7/55 oxidoreductase, acting on a sulfur group of donors

2/31 protein-disulfide reductase

1/24 oxidoreductase, acting on a sulfur group of donors, NAD(P) as acceptor

2/14 metalloaminopeptidase

p < 0.001

p < 0.01

p < 0.05

SUMMER – Molecular Function (MF)